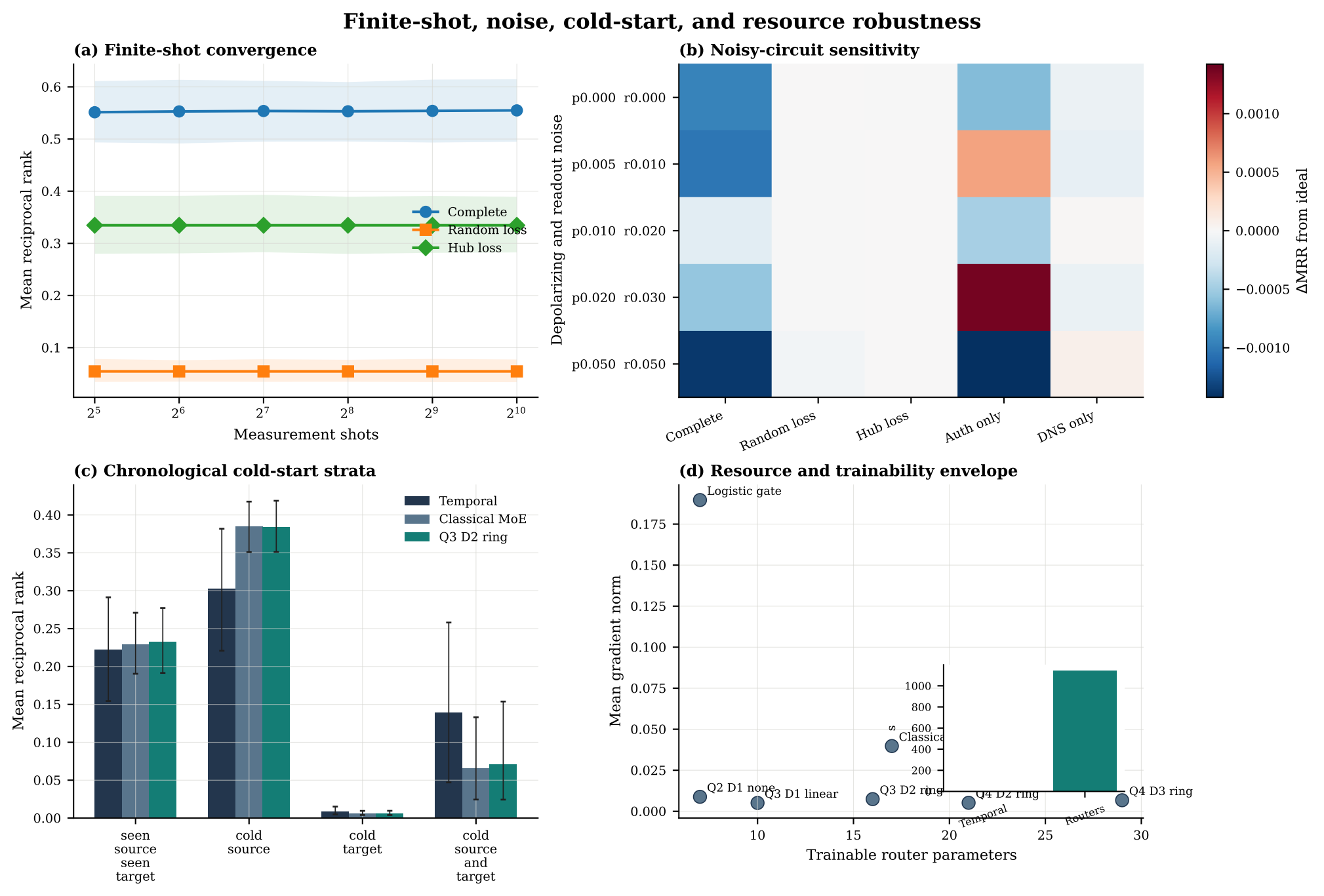
Finite-shot, noise, cold-start, and resource robustness
(a) Finite-shot convergence
Mean reciprocal rank
Measurement shots
Complete
Random loss
Hub loss
(b) Noisy-circuit sensitivity
Depolarizing and readout noise
p0.000 r0.000
p0.005 r0.010
p0.010 r0.020
p0.020 r0.030
p0.050 r0.050
Complete
Random loss
Hub loss
Auth only
DNS only
ΔMRR from ideal
(c) Chronological cold-start strata
Mean reciprocal rank
Temporal
Classical MoE
Q3 D2 ring
seen source seen target
cold source
cold target
cold source and target
(d) Resource and trainability envelope
Mean gradient norm
Trainable router parameters
Logistic gate
Q2 D1 none
Q3 D1 linear
Q3 D2 ring
Q4 D2 ring
Q4 D3 ring
Temporal
Routers